

- Seasons capturing tournament editions
- Venues representing stadiums
These nodes intentionally remain lightweight and descriptive rather than analytical.

Relationship first modelling

The analytical power of the model comes from where the data lives. Instead of embedding performance metrics into nodes, they are attached to relationships where context exists.

Examples include:

- Runs scored and strike rate in the player to match relationship
- Overs bowled and wickets in the bowler to match relationship
- Team membership in a player to team relationship in a season

This approach preserves temporal correctness and prevents historical distortion when players change teams or roles.

Graph scale and infrastructure

The final Neo4j deployment contains more than 320,000 nodes and over 1.2 million relationships spanning multiple IPL seasons. At this scale, the graph reflects real tournament complexity while remaining efficient for deep traversal queries. Indexing and relationship-centric access patterns allow multi-hop queries without the join explosion typical of relational systems.

Once the IPL is represented as a graph, analysis shifts away from ranking tables and towards patterns of interaction.

Player adaptability across contexts

Rather than measuring success purely through aggregate runs or wickets, players can be evaluated on how they perform across varying conditions. Traversing player to team to venue paths reveals who adapts

consistently versus who thrives only in specific environments.

Team cohesion and continuity

Team performance can be analysed through relational stability. Teams that retain a coherent player core across seasons often exhibit stronger structural signals than teams built around frequent overhauls. These patterns are visible in the density and persistence of player team relationships rather than in match results alone.

Identifying structural influencers

Graph algorithms introduce another dimension of insight. Centrality measures highlight players who act as connectors across teams and seasons. These players often influence tournament dynamics in ways not immediately visible through traditional statistics. Community detection

surfaces clusters of players and venues that share similar performance characteristics, offering a data-driven view into tactical groupings.

Venues as contextual forces

Venues in the IPL are often reduced to simple labels, yet their influence is persistent and measurable. When modelled as first class entities, venues reveal how conditions shape performance over time rather than in isolated matches.

Sub-graphs anchored to specific venues expose:

- Players whose performance consistently improves or degrades at certain grounds
- Teams that rely heavily on home conditions for sustained success
- Venue-specific biases towards batting depth or bowling discipline

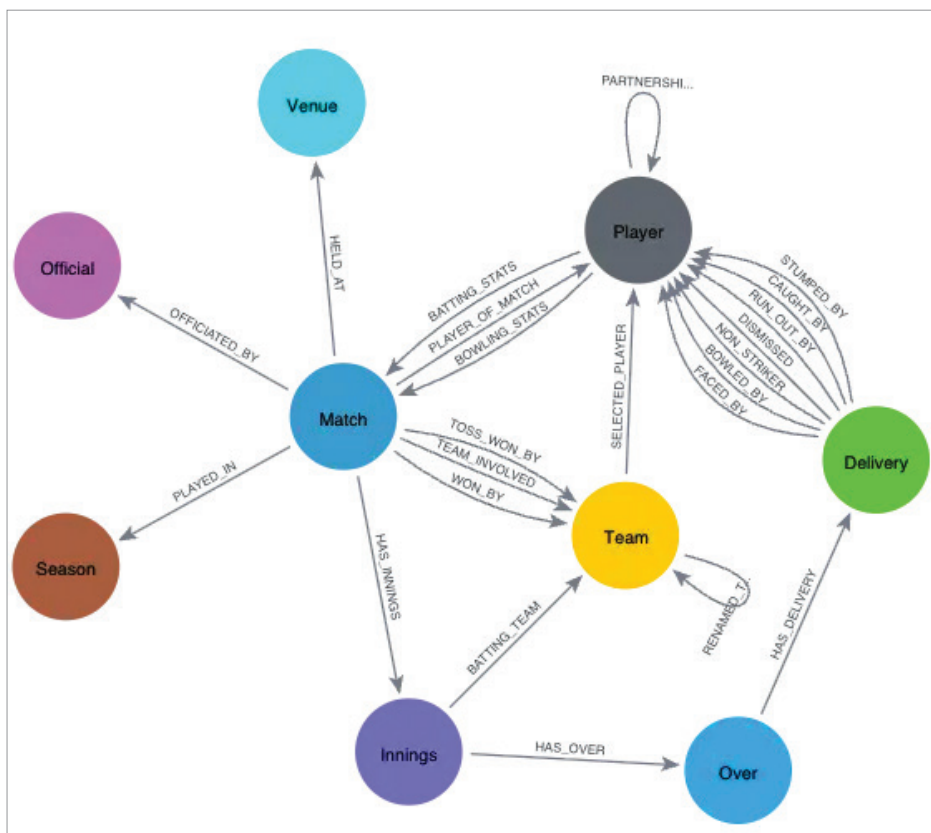


Figure 1: Relationship first modelling